

IBM TechXchange
2026 Pre-
conference Dev
Day Hackathon

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The hackathon expectation

In this hackathon, you will design and build a **working prototype** using **IBM Bob IDE**, aligned to the following theme:

Build with purpose using IBM Bob 2.0

Create a solution that improves a specific developer workflow, such as onboarding, debugging, code review, testing, application maintenance, or release and deployment processes. Start by clearly defining a problem where time, effort, or errors are too high today. Then, using IBM Bob 2.0, build a working prototype on a real or sample project that demonstrates a full solution to improve the specified workflow. Leverage features like Agent mode, parallel tasks, subagents, and document understanding to manage and improve multiple steps, not just assist with coding. Clearly demonstrate impact by showing how your solution increases productivity, reduces manual effort, errors, and rework, or significantly shortens the time required to complete tasks.

Refer to [example use cases](#) to help you brainstorm ideas for building a solution.

Important: Participants are required to upload all relevant **Bob IDE task session summary screenshots** to their code repository as evidence of Bob usage. Learn [how to upload task session summary](#).

Participants may also **optionally** use the following watsonx products:

- [IBM watsonx Orchestrate](#) – a no-code and low-code platform designed to orchestrate AI agents across business workflows. With watsonx Orchestrate, you can create, deploy, and manage intelligent agents and assistants that automate tasks, streamline processes, and enhance productivity.
- [IBM watsonx.ai](#) – a powerful AI studio that supports the development of agentic AI solutions using a wide range of foundation models. Through the Prompt Lab, participants can experiment with IBM’s Granite models and other leading models to create agents that understand and respond to natural language. IBM watsonx.ai can also serve as an inference provider for your agents, allowing them to generate responses, make decisions, and interact with users or systems intelligently.

A note on using other technologies

You may use any framework or technology to build your solution as long as you adhere to the product usage policies. However, to be eligible for judging, your solution must showcase **IBM Bob IDE** as a core component.

A note on data sets before you begin

Participants are required to bring their own datasets to build the solution aligning to your use case. As you collect data for your project, you’ll want to use best practices. Here are helpful tips:

- Teams are responsible for ensuring data is compliant.
- Data from public websites may be used, if the terms allow for commercial use, but please keep a list of the websites you use.
- Do not use data or assets containing company confidential data, or any other data without permission from the data owner. Teams are responsible for getting approval.
- Do not use any client data.
- Do not use any data containing personal information (PI).
- Do not use data obtained from social media.

Get started with IBM Bob

To access and use the hackathon-provisioned IBM Bob account, participants must be registered for the hackathon.

Important: If you already have access to Bob through your personal account, please use the hackathon-provisioned Bob account only for this event and to avoid consuming usage on your personal account.

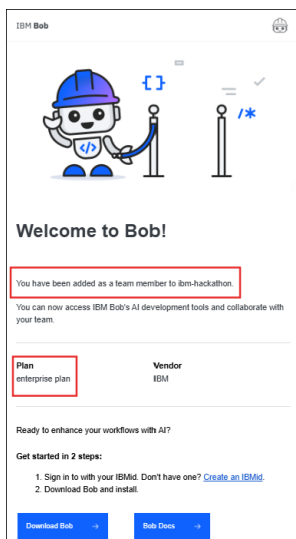
Confirm your hackathon IBM Bob account

At the start of the hackathon, all registered members will receive an email invite to join the **hackathon-**

provisioned Bob account. Follow the steps below to confirm your account:

1. Check the email inbox you used to register for the hackathon and open the email invite you received from the IBM Bob team about joining your hackathon-provisioned Bob account. To confirm you have the correct email, look for:
 - You have been added as a team member to **ibm-hackathon-xxxx**
 - **Note:** The team member reference mentioned in the email does not mean that you have been added to a team for this hackathon. It is simply terminology used as part of the account enablement process through an Enterprise account. Participants who wish to join or form a team must do so through the supported hackathon platforms.
 - **Bob Plan:** Enterprise plan

Please check your Junk/Spam folders if you cannot find the email in your inbox. You can also quickly search for “IBM Bob” to locate the email.



2. [Create an IBMid](#) if you have not created one yet. IBMid is required to access Bob.

Bobcoins

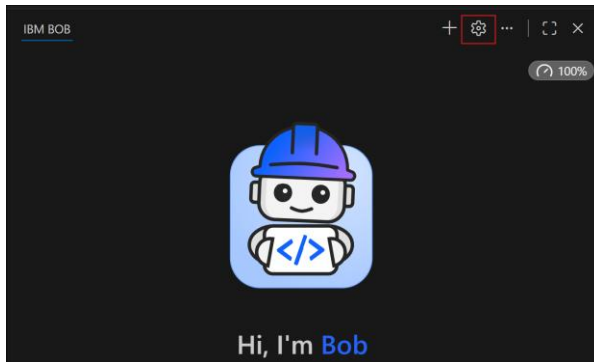
Each interaction with Bob's AI capabilities consumes Bobcoins. For this hackathon, **40 Bobcoins** will be automatically applied to your hackathon-provisioned **IBM Bob account**. These Bobcoins are intended to be sufficient for designing and building a compelling proof-of-concept submission.

It is recommended that you plan ahead with your teammates to divide tasks so that you can make full use of the total Bobcoins allocated across all team members. Once your Bobcoin usage reaches **100% usage, no additional Bobcoins will be provided** as part of the hackathon. You may still continue working on your solution and final submission. For example, consider leveraging the optional watsonx technologies enabled for this hackathon.

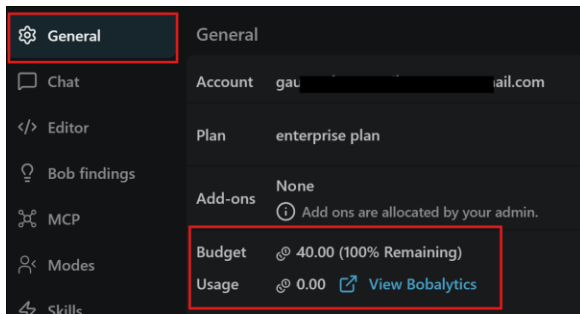
Follow [Bob best practices](#) to use the tool effectively and efficiently.

You can monitor your Bobcoin usage in Bob IDE or [Admin dashboard](#):

- In your Bob IDE, select the **Settings** icon and the **Settings** tab will open.



- Under the **General** section, you can view your Bobcoin usage.



- If you have access to multiple Bob accounts, ensure that you select the hackathon provisioned instance named **ibm- coding-challenge-uat (region: us-east)**.

Note: The team name reference mentioned in the settings does not mean that you have been added to a team for this hackathon. It is simply terminology used as part of the account enablement process through an Enterprise account. Participants who wish to join or form a team must do so through the supported hackathon platforms.

Working with IBM Bob

To participate in the hackathon, you must install and use **Bob IDE**. You may also optionally use **Bob Shell**.

1. [Bob IDE](#) (Required)
2. [Bob Shell](#) (Optional)

Important: You must use **Bob IDE** for this hackathon. As part of your submission, you will be required to upload your [Bob task session summary](#) as evidence of Bob usage.

1. Bob IDE

Install the Bob IDE, sign into your Bob hackathon account, and start building your solution with Bob.

Install Bob IDE

Follow the [Bob IDE installation instructions](#).

Important: IDE v1.0.3 and v2.0.0 will stop working on September 30, 2026.

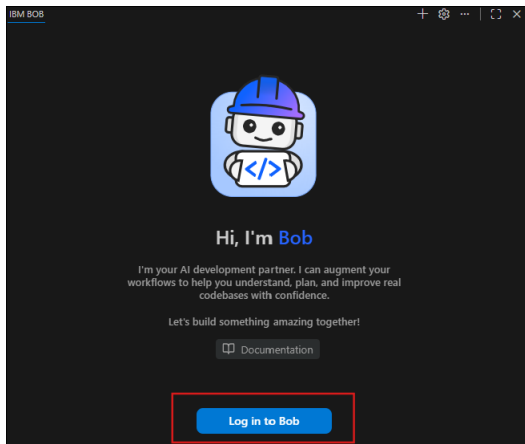
- If you are on v1.0.3, you must upgrade to the latest v2.0.x release.
- If you are on v2.0.0, you must upgrade to v2.0.2 or later.

Sign into Bob IDE

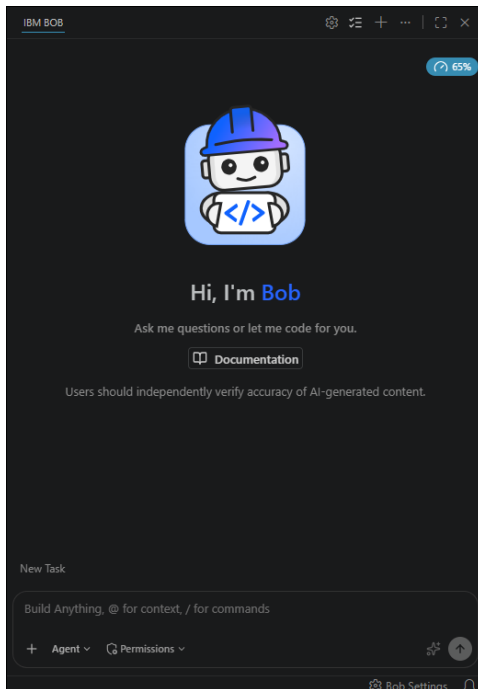
To use Bob IDE after installing it, you have to sign in to Bob using your hackathon registration email.

Important: An IBMid is required to authenticate. If you do not have one, see [Create an IBMid](#) to create one for your hackathon registered email.

- Open your Bob IDE and select the **Log in to Bob** button on the IDE.

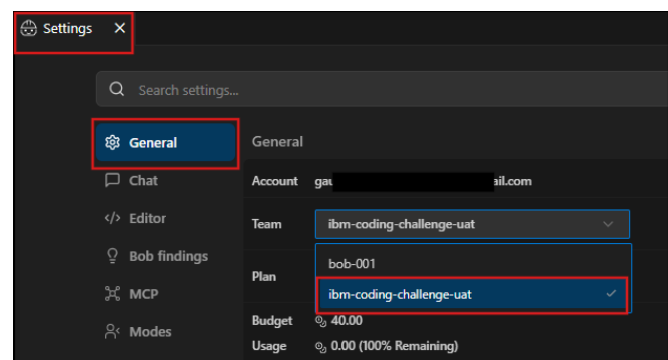
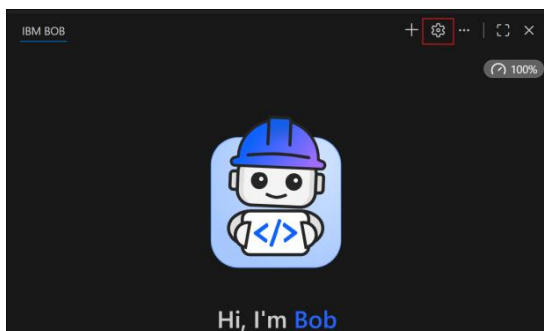


- Follow and complete the IBMid authentication process on your browser. Once authentication is complete, Bob chat interface will be displayed for you to get started.



- **Important:** If you have access to multiple Bob accounts, ensure that you select the hackathon provisioned instance named **ibm- coding-challenge-uat (region: us-east)** to avoid consuming Bobcoins from your personal account. To switch to hackathon-provisioned account, select the **Settings** icon in the Bob IDE and it will open the Settings tab. Under General, you can switch the team to **ibm-coding-challenge- uat (region: us-east)** instance.

Note: The team name reference mentioned in the settings does not mean that you have been added to a team for this hackathon. It is simply terminology used as part of the account enablement process through an Enterprise account. Participants who wish to join or form a team must do so through the supported hackathon platforms.



Note: If you experience network issues with outbound traffic, you might need to configure your firewall settings. For more information, see [Configuring firewall rules for Bob](#).

Best practices

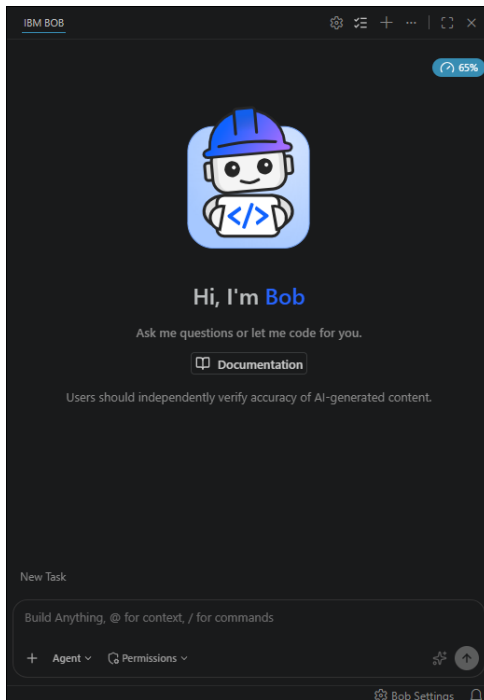
Follow the [best practices guidelines](#) to get the most out of your experience with Bob.

Bob IDE features and configuration

Explore Bob IDE features and their configuration to enhance your experience working with Bob and solution building.

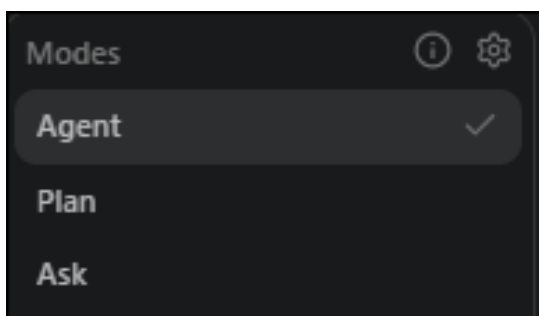
- [Chat interface](#)

Bob IDE's chat interface is your primary workspace where you use natural language prompts, slash commands, and auto-approval for AI-assisted coding workflows. Learn more about [the chat interface and its components](#).



- [Modes](#)

Modes are specialized personas that tailor Bob's behavior for your specific tasks. Each mode offers different capabilities and access levels to help you accomplish particular goals more efficiently. Learn more about [modes and their usage](#).



- [Custom modes](#)

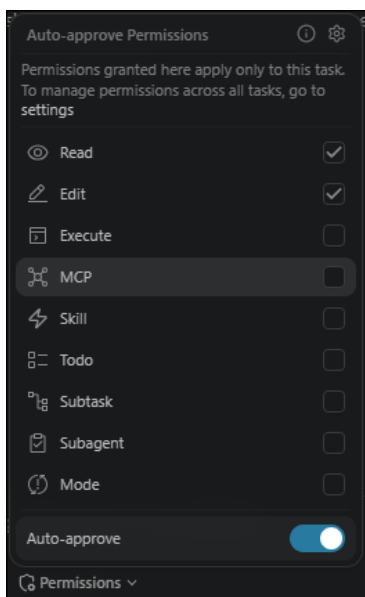
Tailor Bob's behavior for specific tasks or workflows by creating custom modes. You can make modes global (available across all projects) or project-specific. Learn more about [subagents](#).

- [Subagents](#)

Learn how Bob uses subagents to handle focused, self-contained tasks in isolated contexts without polluting the main conversation. Learn more about [custom modes and its configuration](#).

- [Auto-approve](#)

Auto-approve actions in Bob IDE eliminate repetitive prompts but increase security risks. Learn safe configuration practices for read, write, and execute permissions. Learn more about [auto-approve and its configuration](#).

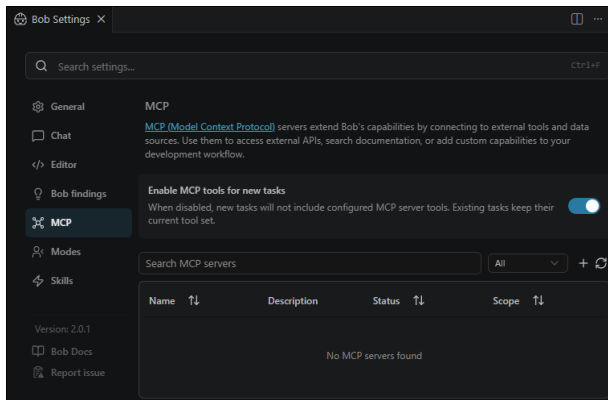


- [Custom rules](#)

Custom rules influence how Bob responds to your requests, aligning output with your specific preferences and project requirements. You can control coding style, documentation approach, and decision-making processes. Learn more about [custom rules and its configuration](#).

- [MCP servers](#)

The Model Context Protocol (MCP) extends Bob's capabilities by connecting to external tools and services. This guide shows you how to configure, manage, and use MCP servers with Bob. Learn more about [MCP server configuration with Bob](#).



- [Ignoring files](#)

Control which files Bob can access by creating a `.bobignore` file in your project. Learn more about [Ignoring files](#).

- [Context mentions](#)

Context mentions let you reference specific elements of your project directly in your conversations with Bob. By using the @ symbol, you can point Bob to files, folders, problems, and other project components, enabling more accurate and efficient assistance. Learn more about [context mentions](#).

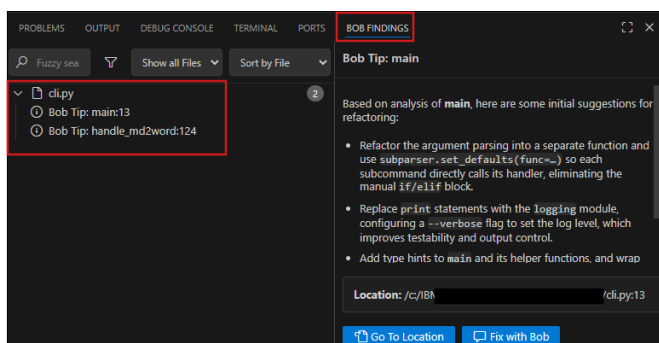
- [Security guidelines](#)

Bob has capabilities for coding and system interaction. To use safely, follow the [security guidelines](#).

```
> .bobignore
# Example .bobignore patterns
.env
secrets/
*.key
config/credentials.json
```

- [Bob tips](#)

Bob tips detects code quality issues in real-time with AI-powered refactoring suggestions for complex functions. Reduce technical debt as you code. Learn more about [Bob tips and findings](#).



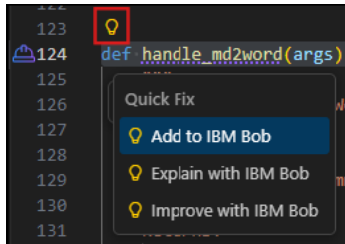
- [Rollback](#)

Automatically version workspace files during AI tasks. Safely explore AI suggestions and experiment

with changes, and easily recover from unwanted changes. Learn more about [Rollback](#).

- [Code actions](#)

Access Bob's AI-powered code actions that provide quick fixes, refactorings, and code explanations directly in the Bob IDE through lightbulb icons and context menus. Learn more about [utilizing code actions](#).



- [Code reviews](#)

Use the built-in Review workflow to get AI-powered code reviews directly in your IDE. Bob analyzes changes and flags potential issues before you commit your work. Learn more about [code reviews and its usage](#).

Quick Reference Card	
Command	Use Case
<code>/review</code>	Quick code check
<code>/review --branch main</code>	Pre-PR review
<code>/review #123 --issue-coverage</code>	Verify issue completion
<code>/review --pr-desc</code>	Generate PR description
<code>/review file.py</code>	Focused file review

- [Commit messages](#)

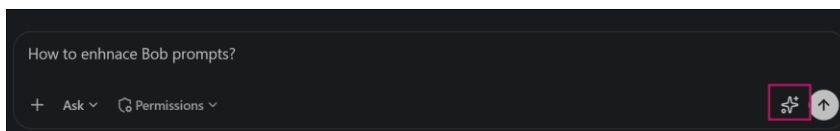
Generate meaningful commit messages automatically with Bob. Save time, maintain consistency, and follow conventional commit standards with AI assistance. Learn more about [commit message and its usage](#).

- [Pull requests](#)

Bob can generate pull requests (PRs) and PR descriptions directly from your IDE, streamlining your development workflow and saving time. Learn more about [pull requests and its usage](#).

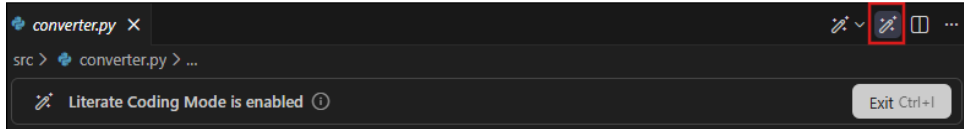
- [Enhance prompt](#)

Improve your prompts before sending them to Bob. Click the sparkles icon in the chat input to automatically refine your request, making it clearer, more specific, and more likely to yield better results. Learn more about [enhancing prompts](#).



- [Literate coding](#)

Write code with AI assistance directly in your editor using natural language instructions right where the code should go. Bob generates the implementation in context with inline diffs. Learn more about [literate coding and its usage](#).



- [Skills](#)

Create reusable instruction sets for specialized workflows. Define custom workflows, add supporting files, and teach Bob specialized tasks for consistent results. Learn more about [skills and its usage](#).

Bob IDE hands-on exercises

Try the quick start hands-on exercises for sample use cases to get started with using Bob.

Important note: You must use the [hackathon-provisioned IBM Bob account](#) to avoid usage on your personal Bob account, if you have one.

- [Bob exercises](#)

Quick hands-on exercises to learn how to use IBM Bob in the IDE for understanding code, refactoring, and automating everyday development tasks.

- [Bob and watsonx Orchestrate exercises](#)

Hands on exercises that show how IBM Bob and watsonx Orchestrate work together to design, build, and run agent-based workflows.

Bob exercises:

- [Quickstart exercise](#)

Use IBM Bob to build a web UI for an existing Node.js Express API and run it in Docker, using modes, the approval workflow, and agentic iteration.

- [Travel demo app](#)

Get started with IBM Bob and the Galaxium Travels demo app by learning the tutorial flow, setup requirements, and application architecture.

- [Build agents with Bob /init](#)

Give Bob persistent project context across conversations and modes with /init, which automatically generates AGENTS.md files.

- [Create a commit and pull request with Bob](#)
Use Bob to create a branch, stage changes, generate a commit message, push to your repository, and open a pull request.
- [Generate code from comments](#)
Use literate coding to have Bob generate code from natural language comments, where Bob helps you write precise, context-aware code modifications directly in your editor.
- [Plan and implement complex features](#)
Learn how to use the agentic chat sidebar to employ a Plan mode and Code mode workflow, where Bob plans complex features end-to-end and then implements them autonomously across your full codebase.
- [Standardize Bob's behavior](#)
Standardize Bob's behavior across your team using project-level rules files that tell Bob to document its code and remember its previous actions.
- [Add a custom mode](#)
Extend Bob's job capabilities by creating a custom product-manager mode with a tailored role definition, behavioral instructions, and deterministic tool access constraints.
- [Create a new context window](#)
Manage Bob's context window to preserve memory, control cost, and maintain output quality during complex or long-running conversations.
- [Modernize a Node.js application](#)
Learn to use IBM Bob for application modernization by upgrading a Node.js Express API from version 16 to 22. Try AI-assisted development with modes, approvals, and literate coding in this hands-on tutorial.
- [Inspect an unfamiliar codebase](#)
Use IBM Bob to rapidly understand an unfamiliar application, such as its purpose, project structure, architecture, tech stack, key components, test coverage, and deployment model. You can do this without relying on outdated documentation or waiting for teammates.
- [Generate architecture diagrams](#)
Use IBM Bob to analyze the Galaxium Travels codebase and generate Mermaid UML class diagrams, sequence diagrams, and use case diagrams. Learn how to use context mentions in Ask mode to explore code and Agent mode to save the results to your repository.
- [Audit code and generate reports](#)
Use IBM Bob to create a reusable security audit skill, scan an application against OWASP ASVS requirements, and generate SARIF and OSCAL reports that developers and AI agents can act on.

- [Generate secure code with an actor-critic workflow](#)

Use IBM Bob to configure security rules and apply an actor-critic pattern to generate Python code that satisfies security frameworks before it reaches a static analysis tool.

Bob and watsonx Orchestrate exercises:

- [Using IBM Bob to build watsonx Orchestrate agents and MCP tools](#)

A hands-on guide for automating the full process of building and deploying an agentic workflow using IBM Bob.

- [Build agentic workflows programmatically on watsonx Orchestrate using IBM Bob](#)

A hands-on guide for creating automated invoice-processing agentic workflows using IBM Bob to generate code, tools, and configuration for watsonx Orchestrate.

- [Turn BPMN diagrams into production-ready agents with Bob skills and watsonx Orchestrate](#)

A hands-on guide for using Bob skills to transform BPMN process models into SOP-driven, fully tested, and deployable watsonx Orchestrate agents with best-practice workflows, tools, and automation patterns.

2. Bob Shell (Optional)

Bob Shell brings IBM Bob's AI capabilities to your command line. Bob Shell delivers the same context awareness and reasoning-focused approach from IBM Bob but optimized for shell environments and automated processes.

Install Bob Shell

Follow the [Bob Shell installation instructions](#).

Important: Upgrading from Bob Shell 1.0.x

Bob Shell 2.0.0 requires a fresh install. Currently there is no automated upgrade path from 1.0.x. Your existing settings and configurations will be preserved during the install.

Authentication

Bob Shell uses the same login entry point as all other Bob clients: bob.ibm.com/login. Learn more about [Bob Shell authentication](#).

Start an interactive session

Interactive sessions provide a conversational interface to Bob directly in your terminal, allowing real-time assistance with your development tasks. Learn more about [starting interactive sessions](#).



Start a non-interactive session

Non-interactive session provides a method to use Bob Shell directly from the command line without entering an interactive session. Use for automation, scripting, and batch processing tasks. Learn more about [starting non-interactive sessions](#).

Bob Shell configuration

You can configure Bob Shell to match your workflow preferences. Learn more about [Bob Shell configuration](#).

Bob Shell tools

Learn how Bob Shell uses specialized tools to read files, edit code, run commands, and interact with your development environment from the command line. Learn more about [Bob Shell tools](#).

Bob Shell features

Explore Bob Shell features to enhance your experience and solution building.

- [Slash commands](#)
Create custom slash commands to automate workflows and standardize team practices. Learn more about [Slash commands](#).
- [Modes](#)
Modes are specialized personas that tailor Bob's behavior for your specific tasks. Each mode offers different capabilities and access levels to help you accomplish particular goals more efficiently. Learn

more about [Modes](#).

Bob Shell usage examples

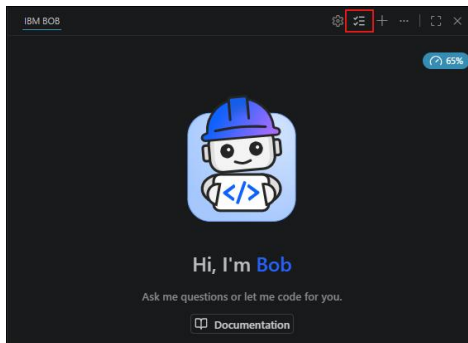
Try practical examples on how to use Bob Shell for debugging, code improvement, file creation, documentation generation, and learning new concepts. Learn more about [Bob Shell usage examples](#).

Uploading Bob task session summary

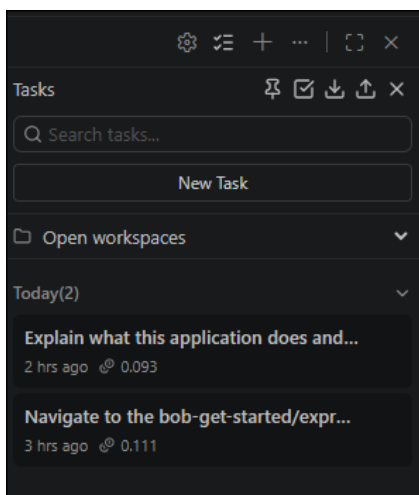
Each team member must upload all relevant Bob task session summary screenshots as part of your project submission. These screenshots must be included in your **final code repository** as **submission deliverable**.

Follow the steps below to complete this process:

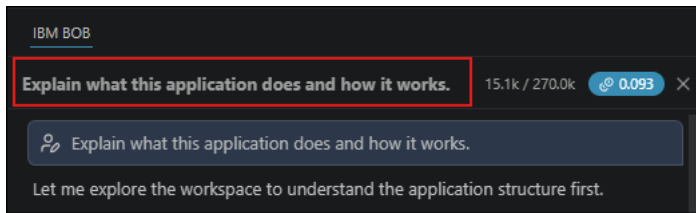
- Create a folder named **bob_sessions** in your project submission code repository.
- In your Bob IDE's chat interface, select **Tasks** to display the task list.



- Select a task from the list related to your project submission and the selected task will open in the chat panel. Confirm that you are in the correct project workspace. If your submission includes tasks from multiple workspaces, select **All** to view tasks across all relevant workspaces.

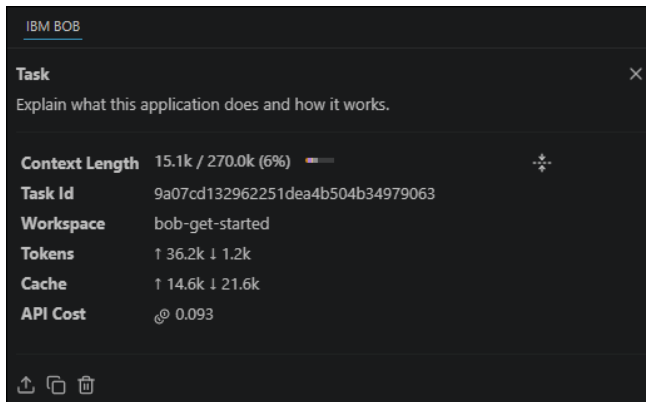


- Select the task header. A task session consumption summary will be displayed.



- Take a screenshot of the task session consumption summary. Save screenshots in **PNG** format where possible for better text clarity. Use a clear file name that includes your team name, task number, and short task description.

Example: teamalpha_task01_login_flow_summary.png



- Now, repeat the steps above for all the tasks related to your project submission.
- Upload all task session consumption summary screenshots to the **bob_sessions** folder in your code repository.

Important: Before submitting your project, ensure that your team final code repository includes the **bob_sessions** folder with all required screenshots relevant to the project solution.

Get started with IBM watsonx (Optional)

To access and use IBM watsonx products for this hackathon, participants must be registered for the hackathon and have access to the [hackathon site](#). Once you have access to the hackathon site, follow the instructions on the “**Complete the hackathon**” page to request a pre-configured IBM Cloud account for your team. This account will provide the necessary environment to work with the supporting IBM products for this hackathon.

Request a team IBM Cloud Account

- If the button to request is grayed out, that means one of your teammates has already requested your team's account. This account is only to use during and for the purposes of this hackathon and will be closed on September 1.
- Once you or a teammate have submitted an account request, **please allow 2 hours for your account to become active**. At that time, all members of the team will receive an email invitation from IBM Cloud to join the account. Check the email you used to register for the hackathon, and check your junk/spam folder if you are not able to find the email in your inbox.
- If after two hours none of your teammates receive this email, or if you have other issues accessing the account, please reach out to the hackathon mentors in Slack for assistance.

Note on available services

The IBM Cloud account provisioned for this hackathon is **pre-configured with only the services required** to complete the hackathon. You will not be able to configure any new service or modify permissions for existing ones due to restricted access. If you notice a permission/access issue for any service or the cloud catalog, then they are not required/available for this hackathon.

The following optional IBM Cloud services are provided for your hackathon solution building:

- [Natural Language Understanding](#)
- [Speech-to-Text](#)
- [Text-to-Speech](#)
- [Cloudant](#)

Note on preventing exposure of IBM Cloud credentials

Hackathon participants may use IBM Cloud credentials to build their solutions that leverage IBM technologies on IBM Cloud. During development, testing, collaboration, or while submitting their final project in a public repository, participants may unintentionally expose these credentials on publicly accessible platforms.

Exposing IBM Cloud API keys or any cloud credentials can lead to unauthorized access, misuse of resources, and account suspension. If any IBM Cloud credential associated with your hackathon account is detected in a public repository or publicly accessible platform:

- The credential will be deactivated immediately.
- Your hackathon cloud account access will be suspended until you:
 1. Remove the exposed credential from all public sources. If you are using GitHub, you can refer to [Removing sensitive data from a repository](#).
 2. Rotate and replace the exposed credential.
 3. After completing these steps, confirm remediation with the hackathon support team and request access to your hackathon cloud account.

Please follow the [Preventing exposure of IBM Cloud credentials in public repositories and platforms guide](#) for best practices to avoid disruptions to your project and ensure the security of your IBM Cloud resources.

Note on IBM Cloud service usage

For this hackathon, **\$80 credits** will be automatically applied to your provisioned **IBM Cloud account**. These credits should be sufficient for designing and creating compelling submissions using the services available on your account.

You will receive periodic email notifications about your **credit consumption** at the following usage levels: **25%**, **50%**, and **80%**. Once you reach **100% usage**, your account will be **suspended**. You can appeal the suspension by completing the form shared in the account suspension notification email.

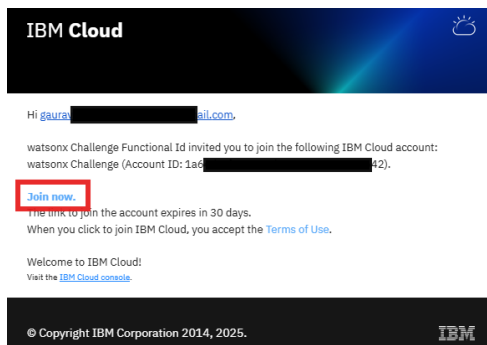
Please note that these email notifications are sent **once per hour**, so there is a possibility that you may **exhaust all your credits before receiving an alert**.

We recommend planning your usage carefully and backing up your work regularly to avoid disruptions.

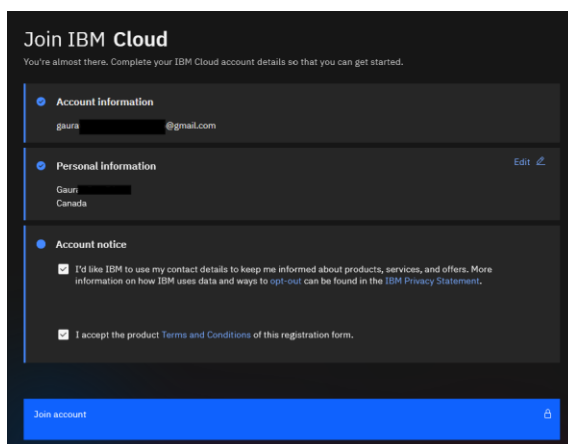
Accessing your hackathon IBM Cloud account

Once your team has been provisioned an IBM Cloud account, all team members will receive an email invite to join the hackathon-provisioned cloud account. Follow the steps below to access your team's cloud account:

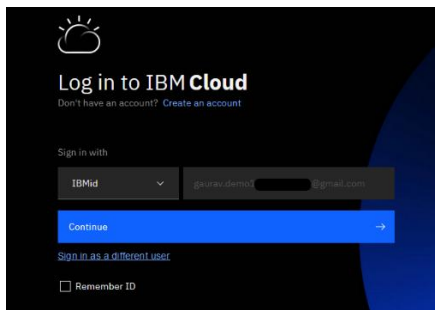
1. Check the email inbox you used to register for the hackathon and open the email you received from the IBM Cloud team about joining your cloud account. Please check your junk/spam folders if you are not able to find the email in your inbox. You can also quickly search for "IBM Cloud" to locate the email.
2. Click the **Join Now** button seen in that email. A new browser tab will open with the cloud account sign up page.



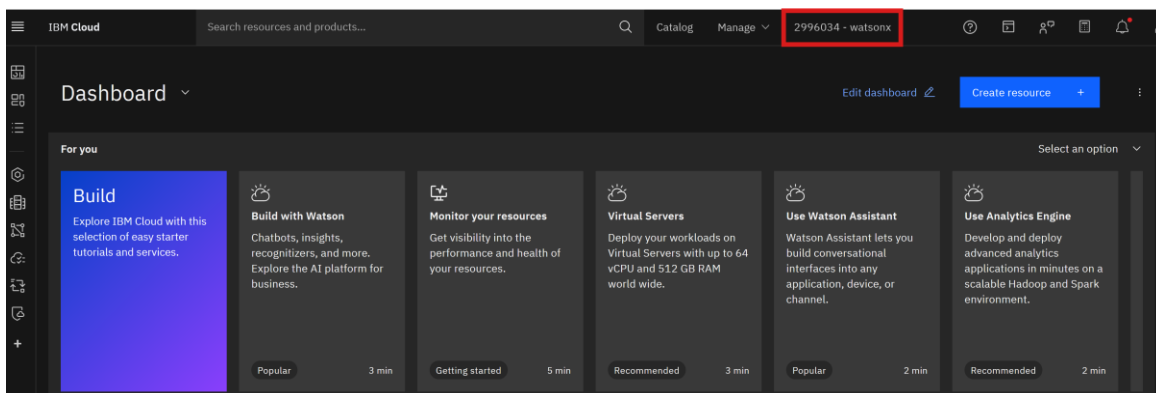
3. Review your account and personal information. Read and accept the Account notice and click the **Join Account** button.



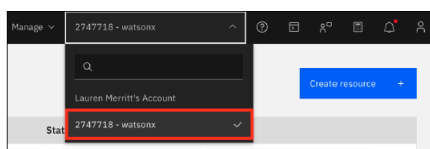
- Complete the authentication process by clicking the Continue button.



- After you authenticate successfully, you will be taken to the IBM Cloud dashboard.



- If you have an existing personal IBM Cloud account for the same email/IBMid, sometimes you will be directed to your personal account. In this case, please switch your account to the **xxxxxxx - watsonx** account. Select your account drop-down at top-right of the dashboard and select watsonx account. Refer to the below image on switching accounts in your cloud dashboard.



Accessing and utilizing IBM watsonx products

To begin building your solution, explore the capabilities and resources for each IBM watsonx product enabled for this hackathon.

- [IBM watsonx Orchestrate \(Optional\)](#)

IBM watsonx Orchestrate is an intuitive, AI-powered platform that you can use to create, configure, and deploy intelligent agents that can automate business tasks. Whether you're automating repetitive workflows or building complex multi-agent systems, the platform is designed to support users of all skill

levels.

2. [IBM watsonx.ai \(Optional\)](#)

A powerful AI studio that enables you to experiment with foundation models like IBM Granite through the Prompt Lab. It can be used to enhance your agents with natural language understanding, intelligent decision-making, and dynamic response generation—serving as an inference engine to make your AI agents smarter and more interactive.

1. IBM watsonx Orchestrate

After successfully [joining the IBM Cloud account](#), you can access watsonx Orchestrate to begin working on the platform and building your solution.

With watsonx Orchestrate, you can:

- Build and deploy agents
- Automate tasks such as scheduling, data entry, and approvals
- Start small and scale up with advanced tools and integrations
- Explore the clean, guided UI that simplifies agent creation and management.

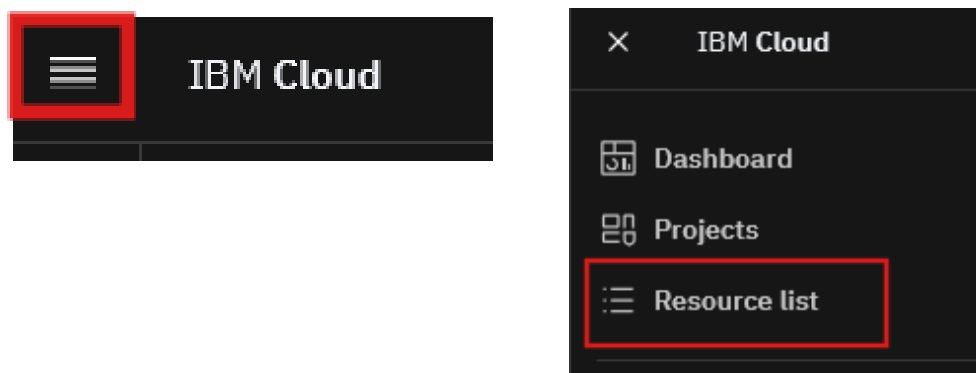
Explore a few [demos](#) on how to use watsonx Orchestrate.

These watsonx Orchestrate features/capabilities are out of scope for this hackathon:

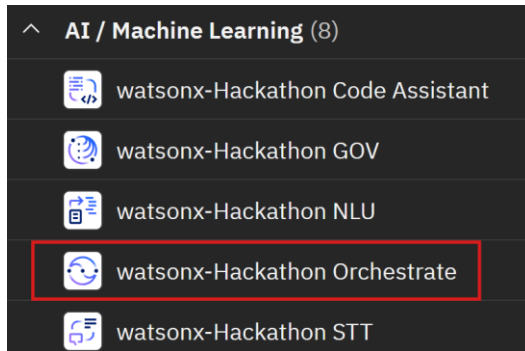
- AgentOps (Preview)

Accessing watsonx Orchestrate

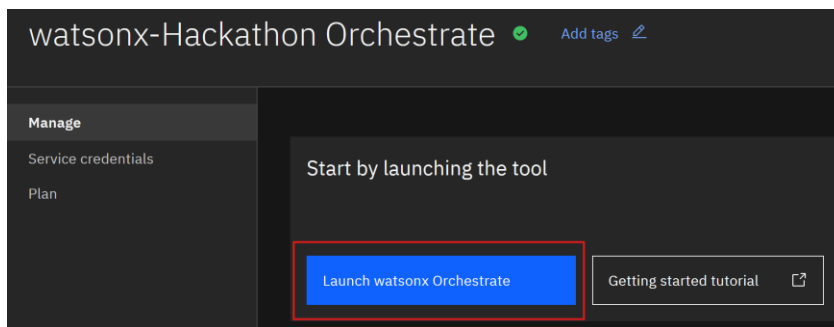
1. In your IBM Cloud account dashboard, select the **Navigation menu** on the top left of the dashboard and select the **Resource list** option.



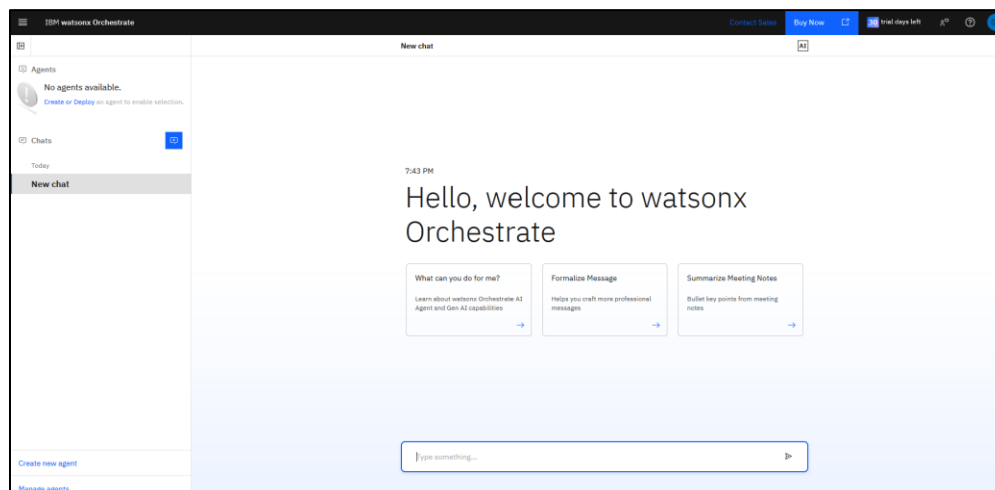
- Expand the **AI / Machine Learning** section and select **watsonx-Hackathon Orchestrate** service.



- You will be navigated to **watsonx-Hackathon Orchestrate** service instance dashboard. Select the **Launch watsonx Orchestrate** button.



- You will be navigated to the watsonx Orchestrate platform with a welcome message and a new chat window.



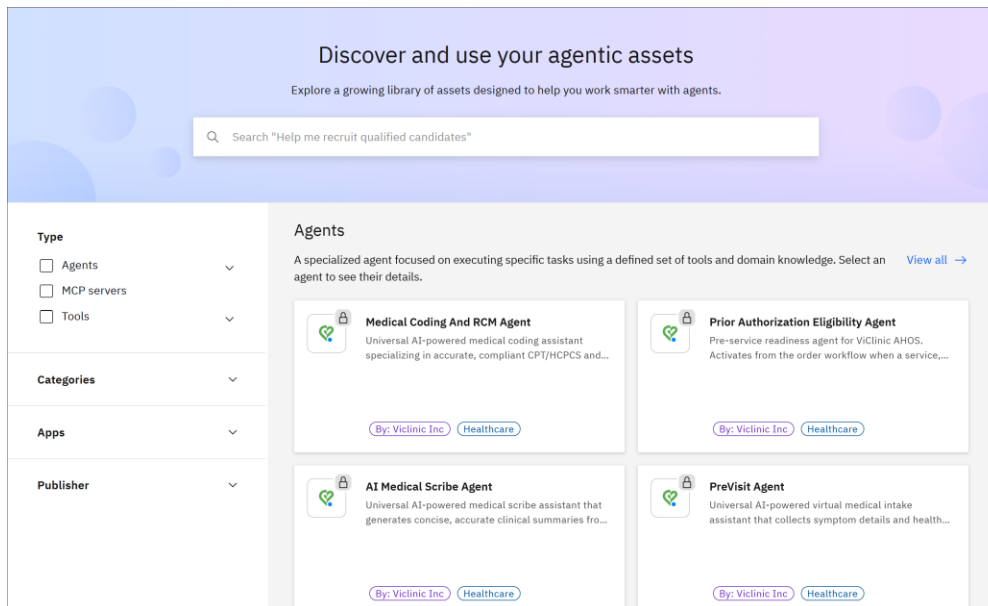
Note: All hackathon participants are assigned the **Builder** role by default, and this role cannot be changed. To learn more about [watsonx Orchestrate user roles](#).

Designing agents

Designing is the first step in building AI agents in IBM watsonx Orchestrate. In this phase, you explore available agents and tools, understand what you can reuse, and prepare to create your own agent when needed.

1. Discovering the catalog

The **Discover** page is your catalog for finding and evaluating prebuilt agents, MCP servers, tools, and apps that you can use to automate work in IBM watsonx Orchestrate. From here, you can search, filter, and browse available assets, then use the selections directly in Orchestrate Chat to complete tasks. Learn more about [discovering the catalog](#).



2. Prepare to build AI agents

Developing intelligent, scalable, and reliable agents in watsonx Orchestrate requires a strategic, multi-phase approach. It goes beyond writing prompts or connecting APIs. Successful agents need thoughtful planning, structured development, testing, and ongoing governance. Learn more about [preparing to build AI agents](#).

Building agents

Building is the core stage in the Agent Development Life Cycle (ADLC) where you create agents, tools, and assistants by defining their logic, behaviors, and interactions. In this phase, you convert design intent into concrete capabilities that support your intended use cases.

In IBM watsonx Orchestrate, building involves creating the components that enable AI-powered automation and assistance.

By building agents, you can:

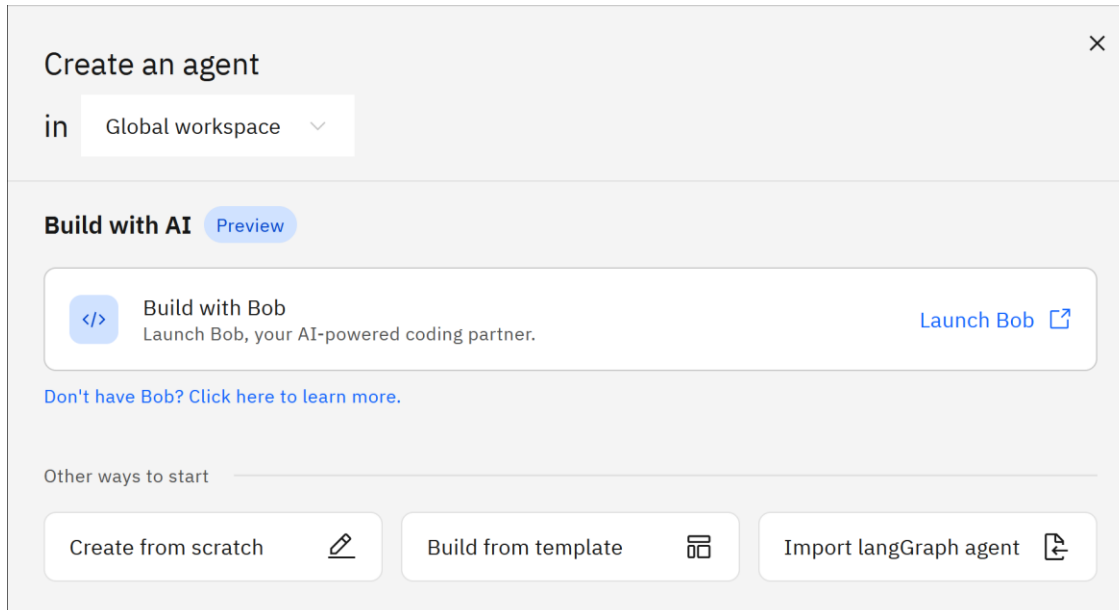
- Build agents that reason, decide, and act autonomously
- Create tools that extend agent capabilities
- Design assistants with conversational interfaces
- Configure knowledge sources and integrations

1. Getting started with building agents

Build your first working agent in watsonx Orchestrate by following a guided path that covers creating, configuring, testing, and publishing an agent. Follow the [essential steps to build your first working agent](#).

2. Creating and customizing agents

Creating and customizing an AI agent involves defining its purpose and personality through thoughtful descriptions and instructions. The **description** defines the agent's purpose and is used by orchestration systems to route requests to the right agent. The **instructions** define the agent's personality and tone, shaping how it responds and engages with users. You can also select an AI model and configure settings such as welcome messages, starter prompts, and voice interactions to create an effective user experience. Learn [how to create an agent with AI or from scratch or from a template](#).



3. Orchestrating agents

Agent orchestration enables you to build scalable, multi-agent systems where a primary agent delegates work to one or more collaborator agents, each focused on a specific domain to complete tasks. Instead of building a single agent that tries to handle every step, you create a system of specialized agents that work

together through structured delegation. Learn more about [orchestrating agents](#).

4. Building agents using the ADK

You can build powerful, customizable agents using the IBM watsonx Orchestrate Agent Development Kit (ADK). The ADK empowers developers to build, test, and deploy agents tailored to specific business needs, all within the watsonx Orchestrate platform. It supports modular design, multi-agent orchestration, and integration with external tools and frameworks. Learn more on [how to use the ADK](#).

5. Building AI assistants

In watsonx Orchestrate, you build the assistant by using Assistant builder. Assistant builder is a chat interface builder that helps to deploy an engaging and embedded chatbot experience. Assistant builder integrates the power of large language models (LLMs) and conversational capabilities of watsonx Assistant to enable responsive and interactive conversation between the users and watsonx Orchestrate.

To learn more, see [Building assistants in AI Assistant builder](#).

Configuring agents

Set up your agent to respond accurately, use the right tools, and access the knowledge it needs. Configuration determines how your agent behaves, what information it can access, and how it interacts with users. A well-configured agent delivers accurate responses, performs appropriate actions, and provides a consistent experience. Learn more about [configuring agents](#).

Building tools

Tools act as interfaces to external capabilities, enabling agents to interact with systems, retrieve data, run calculations, or trigger workflows that go beyond their built-in reasoning. You can build new tools and edit existing tools for your agents to use.

Following are the different ways to add a tool:

- [Agentic workflows](#)
- [Open API tools](#)
- [MCP servers](#)
- [Customizing input-output schemas of prebuilt tools](#)

Testing agents

Test and debug your agent to validate its behavior, identify issues, and improve performance before deployment. Testing and debugging help you verify that your agent responds as expected, uses the appropriate tools, and completes user tasks successfully. Use debugging to inspect how your agent processes individual requests. Use evaluations to measure performance across realistic user scenarios and identify opportunities for improvement. Learn more about [testing agents](#).

Deploying agents

Deploy your Watson Orchestrate agents to production and make them available to users. Deployment is the process of making your agent available to users in a production environment. A successful deployment ensures your agent is accessible, performs reliably, and delivers value to your organization. Learn more about [deploying agents](#).

Using Orchestrate chat

The Orchestrate Chat page is your main workspace for interacting with AI agents, running tasks, and managing conversations. From here, you can start new chats, explore what each agent can do, and complete actions quickly - all with a simple, intuitive interface. Learn more about [using watsonx Orchestrate chat](#).

Operating with agents

Operating is where you manage, monitor, and optimize AI agents in IBM watsonx Orchestrate to ensure they run reliably, perform effectively, and deliver expected outcomes. In watsonx Orchestrate, operating focuses on keeping your agents running smoothly and efficiently. Learn more about [operating with agents](#).

watsonx Orchestrate API

IBM watsonx Orchestrate provides a set of APIs to boost your experience using the features from the product.

- [Getting started with the API](#)
Begin using the API with practical, step-by-step guidance.
- [API reference](#)
Get the methods that you can use to call custom projects, or other resources from the AI chat.

Quick start hands-on exercises

Try the quick start hands-on exercises for sample use cases to get started with using watsonx Orchestrate:

Important notes:

- You must use the hackathon-provisioned IBM Cloud account to access and use watsonx Orchestrate platform to try the below sample exercises.

watsonx Orchestrate SaaS sample exercises:

- [Develop agents with no code using watsonx Orchestrate](#)
A hands-on guide for developing a multi-agent system in watsonx Orchestrate with no code that can communicate with the agents on various channels.
- [Creating intelligent, reusable agentic workflows on watsonx Orchestrate with no code](#)
A hands-on guide for creating AI-driven, human-in-the-loop workflows without writing code.
- [Build context-aware AI agents with watsonx Orchestrate and Astra DB](#)
AI agents are only as effective as their data, making precise retrieval essential for reliable results.
- [AgentOps in watsonx Orchestrate: Observability for Agents with Langfuse and IBM Telemetry](#)
A developer's guide for instrumenting and observing Agents with Langfuse and IBM Telemetry in watsonx Orchestrate.
- [Building and deploying prebuilt domain agents in watsonx Orchestrate](#)
Explore how to work with the prebuilt domain agents and integrate them into existing workflows.
- [Build a scheduled agentic workflow for daily notifications](#)
A practical example of building, importing, and scheduling an agentic workflow that automatically sends daily email notifications using watsonx Orchestrate.
- [Multi-agent orchestration with watsonx Orchestrate](#)
A hands-on journey to design, extend, and integrate AI agents by using Langflow, IBM Granite models, Astra DB, the AI Gateway, and a custom user interface on watsonx Orchestrate.

watsonx Orchestrate Agent Development Kit (ADK) sample exercises:

- [Getting Started with watsonx Orchestrate Agent Development Kit](#)
IBM watsonx Orchestrate includes the Agent Development Kit (ADK) which is a set of developer-focused tools for building, testing, and managing agents. With the ADK, developers get the freedom and control to design powerful agents using a lightweight framework and a simple CLI. You can define agents in clear YAML or JSON files, create custom Python tools, and manage the entire agent lifecycle with just a few commands.
- [Build memory-enabled AI agents in watsonx Orchestrate](#)
Use watsonx Orchestrate and the Agentic ADK to build agents that store user preferences and deliver personalized recommendations across conversations
- [Build an AI agent with Langflow, Granite 4.0 models, and watsonx Orchestrate](#)
A hands-on guide for creating Langflow tools, consuming them with watsonx Orchestrate Agent Development Kit, and using IBM Granite 4.0 Micro model for inference.
- [Connecting to MCP tools with watsonx Orchestrate](#)
A hands-on guide for creating MCP tools and consuming them with watsonx Orchestrate Agent Development Kit.
- [Extend Your AI Agents with External LLMs Using watsonx Orchestrate and AI Gateway](#)
A hands-on guide to bring your own model (BYOM) into watsonx Orchestrate using AI Gateway.
- [Create consistent evaluation workflows for AI agents](#)
A hands-on guide to setup and use a structured evaluation strategy to test, benchmark, and improve AI agent performance.

watsonx Orchestrate and IBM Bob sample exercises:

- [Build agentic workflows with watsonx Orchestrate and IBM Bob](#)
A hands-on guide to automate invoice processing with AI-powered workflows using IBM Bob to generate code, tools, and configuration for watsonx Orchestrate.
- [Using IBM Bob to build watsonx Orchestrate agents and MCP tools](#)
A hands-on guide for automating the full process of building and deploying an agentic workflow using IBM Bob.
- [Turn BPMN diagrams into production-ready agents with Bob skills and watsonx Orchestrate](#)
A hands-on guide for using Bob skills to transform BPMN process models into SOP-driven, fully tested, and deployable watsonx Orchestrate agents with best-practice workflows, tools, and automation patterns.

2. IBM watsonx.ai

After successfully [joining the IBM Cloud account](#), you can access watsonx.ai to begin working on the platform and building your solution.

Note on IBM watsonx.ai service usage

For this hackathon, **\$80 credits** will be automatically applied for the IBM services provisioned on the cloud account. This should be sufficient for designing and creating compelling submissions.

You will receive periodic email notifications about your **credit consumption** at the following usage levels: **25%**, **50%**, and **80%**. Once you reach **100% usage**, your account will be **suspended**. You can appeal the suspension by completing the form shared in the account suspension notification email.

Please note that these email notifications are sent **once per hour**, so there is a possibility that you may **exhaust all your credits before receiving an alert**.

IBM watsonx.ai service consumes the credits when used. Please plan to use it efficiently and back up your work accordingly. Refer [tips to work efficiently on watsonx.ai platform \(Tokens and CUH explained\)](#) and [saving your work](#).

Important:

- **Foundation model inferencing** consumes tokens, which are measured as Resource Units (RUs). **1,000 tokens = 1 RU**, and each RU costs **\$0.0001 USD**.
[Learn more about tokens and tokenization](#).
- If you are using **Jupyter Notebook editor on watsonx.ai**, consider selecting a **lower runtime environment** to avoid high resource consumption and quickly depleting your credits. Notebook runtimes are billed based on **Capacity Unit Hours (CUH)** at a rate of **\$1.02 USD per CUH**.
[Learn more about capacity unit hours and watsonx.ai Studio pricing plans](#).

Note on available watsonx.ai capabilities

The watsonx.ai platform is **pre-configured with only the services required** to complete the hackathon. If you notice a permission/access issue for any service or the cloud catalog, then they are not required/available for this hackathon.

These features/capabilities are out of scope for this hackathon:

- Agent Lab (Beta)
- Bring your own model

- Fine tuning models
- AutoAI pipeline
- AI governance
- Evaluation Studio
- SPSS Modeler

DO NOT USE the below listed watsonx.ai models as they are out of scope for the hackathon and can negatively impact the judgment of your project submission.

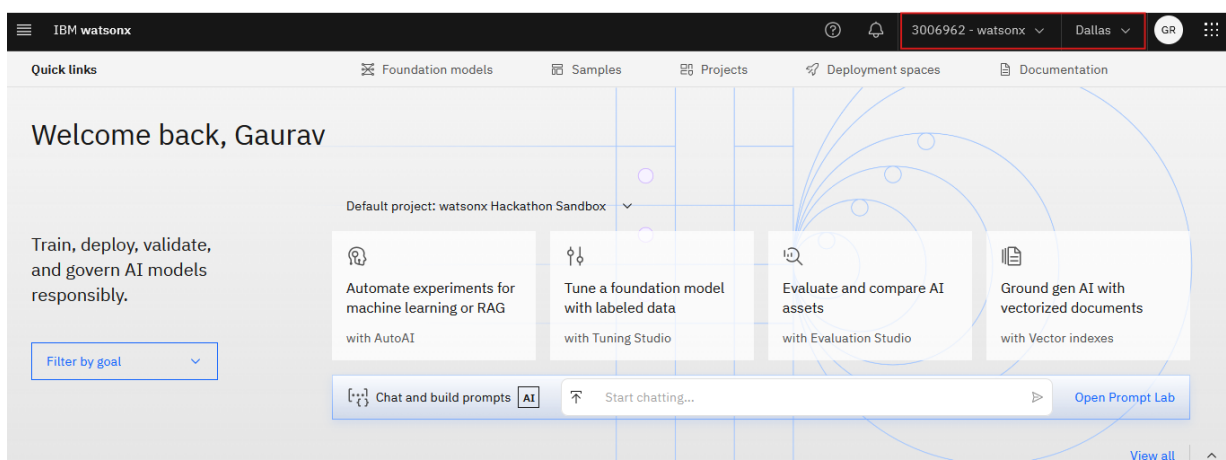
- llama-3-405b-instruct
- mistral-medium-2505
- mistral-small-3-1-24b-instruct-2503

The hackathon-provisioned IBM Cloud account will be deactivated after the completion of the hackathon. Please plan to [save your work](#) at the end of the hackathon.

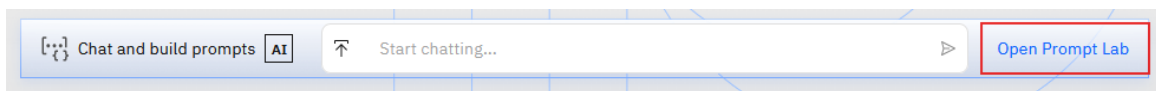
Access Prompt Lab on watsonx.ai

After successfully joining the IBM Cloud account, you can now access the Prompt Lab on watsonx.ai platform to work with the AI models supported on the platform and build your solution.

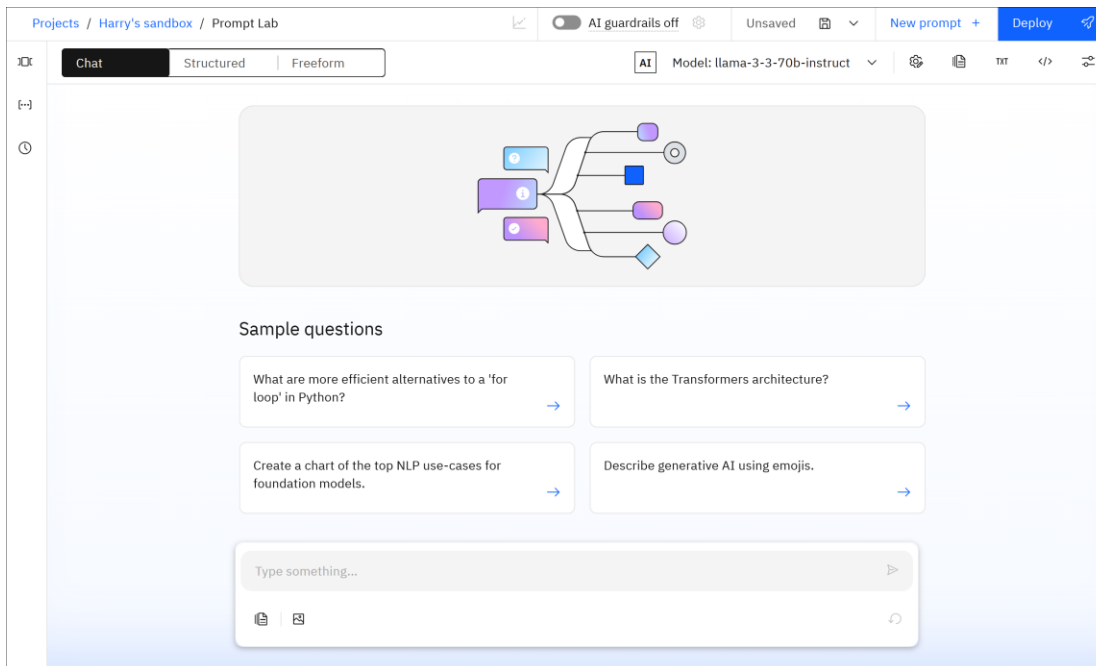
1. Log in to the [watsonx.ai platform](#) with the email you used to access your IBM Cloud account.
2. After successful authentication, you will see “Welcome to watsonx” widget. You can either take the tour or skip it.
3. Next, you will see the watsonx.ai dashboard. Ensure the name of the account is “xxxxxxx – watsonx” and the region is “**Dallas**”.



4. Select the **Open Prompt Lab** button on the “Chat and build prompts” widget.



5. The “Welcome to Prompt Lab” tour will be displayed. You can take the tour to get a quick introduction or skip it.
6. The Prompt Lab Editor opens with a chat window to get you started with the prompt session.



Work with the watsonx.ai Prompt Lab

The watsonx.ai Prompt Lab is an easy-to-use prompt engineering interface where you can experiment with prompting different AI foundation models, explore sample prompts, tune model parameters, integrate applications with an API endpoint, and save and share your best prompts.

[Take a tour of the Prompt Lab](#) and try the [interactive demo](#).

You can access and use the AI models to build your innovative solution using Prompt Lab.

[Learn more about Prompt Lab and how to use it effectively](#).

Prompt Lab editor

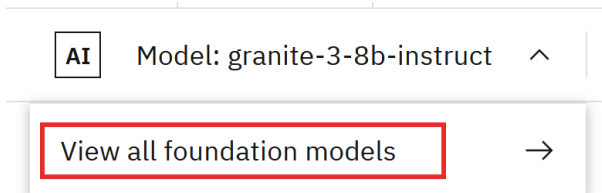
In the Prompt Lab, you can experiment with prompting different foundation models, explore sample prompts, as well as save and share your best prompts. The Prompt Lab editor is a great place to experiment and iterate with your prompts.

However, you can also prompt foundation models in watsonx.ai programmatically. Refer to “[Programmatic access \(API/SDK\)](#)” section.

Selecting an AI model

A default AI model will be pre-selected in the Prompt Lab editor. You can either use the same model or change to a different model. To select a different model:

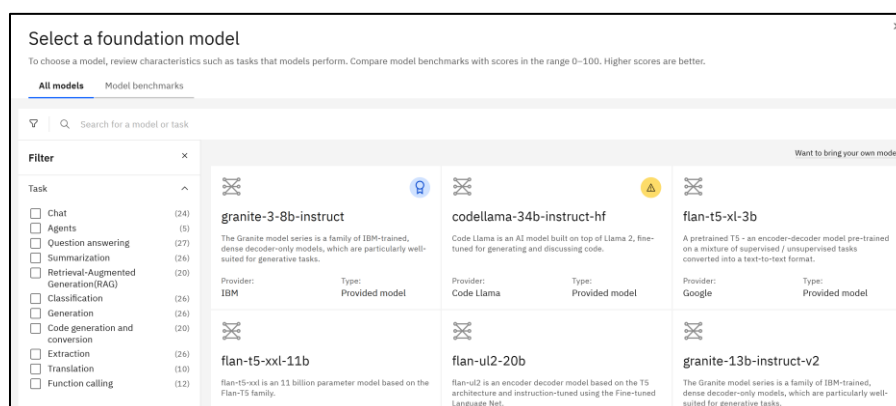
1. Select the AI Model drop-down menu at the top-right of the editor and select **View all foundation models**.



2. A foundation model selection widget appears. Clear the filters to see all the available models. You can use the filters to choose the right model for your solution building. You can select a model tile to learn about the model and use it. If you are limited to only “Chat” supported models, change the Prompt Lab editor to [Structured](#) or [Freeform](#) view and try selecting the models to see all the available model options.

Important: DO NOT USE the below listed watsonx.ai models as they are out of scope for the hackathon and can negatively impact the judgment of your project submission.

- mistral-small-3-1-24b-instruct-2503



To understand how models can address your use case, including information on model modalities, supported languages, tuning, and indemnification, see our product documentation on [choosing a model](#).

Note: Bigger models are not always better. [Learn](#) why smaller models can be better and more cost effective.

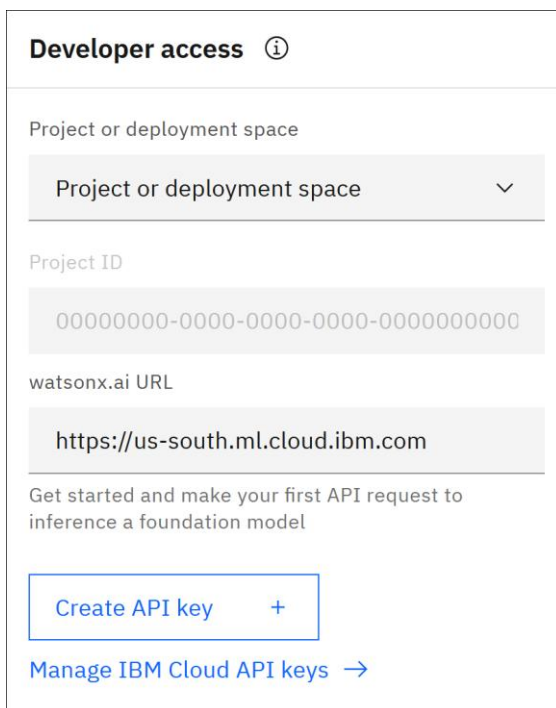
Programmatic access (API/SDK)

You can inference the watsonx.ai models with API or SDK requests.

Developer access information

To use the supported watsonx.ai APIs/SDKs, you will need three values: a **project ID**, an **endpoint URL** and an **API key**.

- Go to [watsonx.ai home page](#).
- Scroll down to the “**Developer access**” section.



Developer access ⓘ

Project or deployment space

Project or deployment space ▼

Project ID

00000000-0000-0000-0000-0000000000

watsonx.ai URL

https://us-south.ml.cloud.ibm.com

Get started and make your first API request to inference a foundation model

Create API key +

[Manage IBM Cloud API keys](#) →

- Select the **Project or deployment space** drop-down and choose the **watsonx Hackathon Sandbox** option if you are working with that project or select any other listed project that you are working with. A **project ID** will be displayed. A default **watsonx.ai endpoint URL** will be displayed for the Dallas region. Ensure the region is always set to **Dallas** at the top right of the watsonx.ai home page.

Developer access ⓘ

Project or deployment space
 watsonx Hackathon Sandbox ▼

Project ID
 c78 [REDACTED] 1456

watsonx.ai URL
 https://us-south.ml.cloud.ibm.com

Get started and make your first API request to inference a foundation model

[Create API key](#) +

[Manage IBM Cloud API keys](#) →

- Select the **Create API key** button. A **Create API key** widget will be displayed. Enter a name, provide optional description and choose the **Delete the leaked key** option. Click the **Create** button.

Create API key ×

You need an IBM Cloud API key to create the IAM bearer token for authenticating your API requests.
[Learn more](#)

Name
 watsonx API key

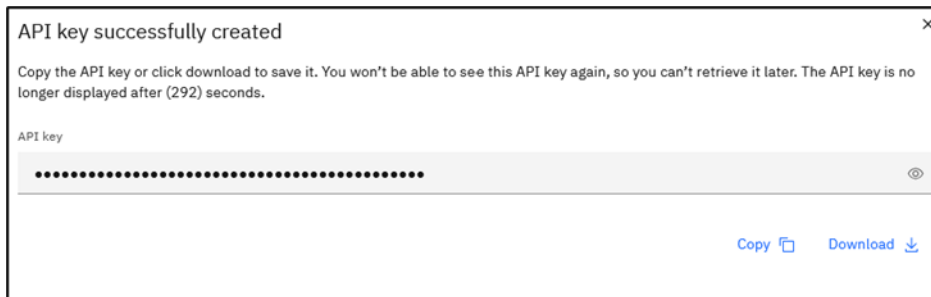
Description (optional)
 Enter name

Choose what to do if this key is leaked:

☐ Disable the leaked key
☒ Delete the leaked key
☐ Nothing

Cancel Create

- An API key will be created successfully. Copy the API key and save it safely to use for calling the API/SDK. You can also download and save the file in a secure path in your system.



watsonx.ai programmatic options

There are multiple options to help you get started using watsonx.ai APIs/SDKs.

Option 1: Prompt Code on Prompt Lab

Refer to the [access prompt code instructions](#) to learn how to quickly get access to the text generation API within the watsonx.ai Prompt Lab.

Option 2: Different watsonx.ai API capabilities

Explore and leverage different watsonx.ai API capabilities in your solution.

- [Chat](#)
- [Agent-driven chat](#)
- [Tool calling](#)
- [Text generation](#)
- [Time series](#)
- [Text rerank](#)
- [Embeddings](#)
- [Text extraction](#)

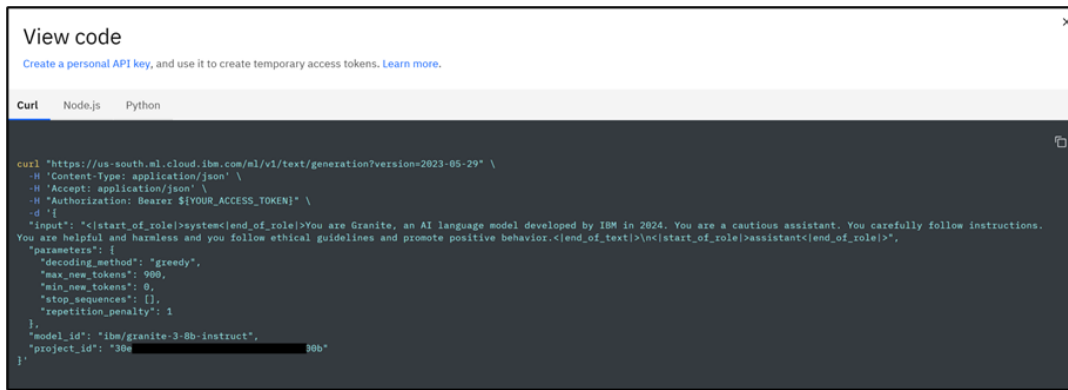
Refer to supported API functionality by AI model [here](#).

Access the prompt code (API) from Prompt Lab editor

To prompt an AI model programmatically, you can view and copy the prompt code by selecting the **View code** icon `</>` at the top-right of the prompt lab editor.



The prompt code is available as a Curl, Node.js and Python.



```

View code
Create a personal API key, and use it to create temporary access tokens. Learn more.

Curl Node.js Python

curl "https://us-south.ml.cloud.ibm.com/ml/v1/text/generation?version=2023-05-29" \
-H 'Content-Type: application/json' \
-H 'Accept: application/json' \
-H 'Authorization: Bearer ${YOUR_ACCESS_TOKEN}' \
-d '{
  "input": "<start_of_role>system<end_of_role>You are Granite, an AI language model developed by IBM in 2024. You are a cautious assistant. You carefully follow instructions. You are helpful and harmless and you follow ethical guidelines and promote positive behavior.<end_of_text>\n<start_of_role>assistant<end_of_role>",
  "parameters": {
    "decoding_method": "greedy",
    "max_new_tokens": 900,
    "min_new_tokens": 0,
    "stop_sequences": [],
    "repetition_penalty": 1
  },
  "model_id": "ibm/granite-3-8b-instruct",
  "project_id": "30a"
}'

```

You will require an IAM access token to authorize the prompt code and need to replace **`${YOUR_ACCESS_TOKEN}`** placeholder in the prompt code. You can create an IAM access token using an API key.

- **API key:**

Refer to [Developer access information](#) to get an API key.

- **Generate IAM Access Token:**

Programmatically generate an IAM access token with the API key using the following cURL command:

```
curl -X POST 'https://iam.cloud.ibm.com/identity/token' -H 'Content-Type: application/x-www-form-urlencoded' -d 'grant_type=urn:ibm:params:oauth:grant-type:apikey&apikey=MY_APIKEY'
```

- **curl -X POST** → Specifies an HTTP **POST** request.
- **URL ("https://iam.cloud.ibm.com/identity/token")** → The endpoint to request an authentication token from IBM Cloud.
- **-H "Content-Type: application/x-www-form-urlencoded"** → Sets the request header to indicate that the data is sent in form-encoded format.
- **-d (Data Payload)** → Sends the required data:
- **grant_type=urn:ibm:params:oauth:grant-type:apikey** → Specifies the OAuth grant type as API Key.
- **apikey=MY_IBM_CLOUD_API_KEY** → Replace MY_IBM_CLOUD_API_KEY with your actual IBM Cloud API key.

Expected Response:

```

{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXLTJ5IiwiaWF0IjE4NzI1ODM1M30",
  "refresh_token": "not_supported",
  "token_type": "Bearer",
  "expires_in": 3600,
  "expiration": 1473188353,
  "scope": "ibm openid"
}

```

Note: An IAM token is valid for up to 60 minutes, and it is subject to change. When a token expires, you must generate a new one. Use the property “*expires_in*” for the expiration of the IAM token that you have just created.

watsonx.ai AI agent libraries and tutorials

Explore the watsonx.ai supported AI agent framework libraries and tutorials to help you get started building your AI agent solution.

- [LangChain](#)
- [LangGraph](#)
- [LlamaIndex](#)
- [CrewAI](#)
- [BeeAI](#)
- [AutoGen](#)
- [Python SDK](#)
- [Node.js SDK](#)

Quick start hands-on exercises

Try the quick start exercises and notebooks for sample use cases to get started with using watsonx.ai.

Important notes:

- Refer to [developer access information](#) section to use watsonx.ai credentials as you try the exercises.
- Some of the exercises could include the usage of old model version. You can replace them with newer versions for better performance and output. To check the latest supported AI models on watsonx.ai, either follow [selecting an AI model on Prompt Lab](#) or refer to [supported foundation models on watsonx.ai](#).
- The hackathon-provisioned cloud accounts **do not support solution deployment**. You can run your solution deployment locally on your machine and showcase them in your submissions.
- **Foundation model inferencing** consumes tokens, which are measured as Resource Units (RUs). **1,000 tokens = 1 RU**, and each RU costs **\$0.0001 USD**. [Learn more about tokens and tokenization](#).
- If you are using [Jupyter Notebook editor on watsonx.ai](#), consider selecting a **lower runtime environment** to avoid high resource consumption and quickly depleting your credits. Notebook runtimes are billed based on **Capacity Unit Hours (CUH)** at a rate of **\$1.02 USD per CUH**. [Learn more about capacity unit hours and watsonx.ai Studio pricing plans](#).

watsonx.ai Prompt Lab app templates:

- [LangGraph LLM app template with function calling capabilities](#) (base template)

- [LlamaIndex Workflow LLM app template with function calling capabilities](#) (base template)
- [CrewAI LLM app template with function calling capabilities](#) (base template)
- [arXiv Research agent](#) (community template)
- [Agentic RAG LangGraph template](#) (community template)

BeeAI Agent Framework:

- [BeeAI framework examples](#)

LangChain and LangGraph:

- [Create a LangChain AI Agent in Python using watsonx](#)
- [Build a RAG agent using LangGraph to answer complex questions](#)
- [Build a LangChain agentic RAG system using the Granite model in watsonx.ai](#)
- [Use watsonx, and LangChain Agents to perform sequence of actions](#)
- [Use watsonx, and LangChain to make a series of calls to a language model](#)
- [arXiv Research agent](#)
- [Base LangGraph LLM app template with function calling capabilities](#)

LlamaIndex:

- [Use watsonx and LlamaIndex for Text-to-SQL task](#)
- [Use watsonx, and `llama-3-1-70b-instruct` and LlamaIndex to make simple chat conversation and tool calls](#)
- [LlamaIndex Workflow LLM app template with function calling capabilities](#)

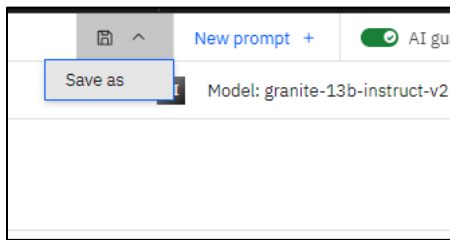
CrewAI:

- [Leveraging CrewAI and IBM watsonx](#)
- [Build an agentic framework with CrewAI memory, i18n, and IBM watsonx.ai](#)
- [Base CrewAI LLM app template with function calling capabilities](#)

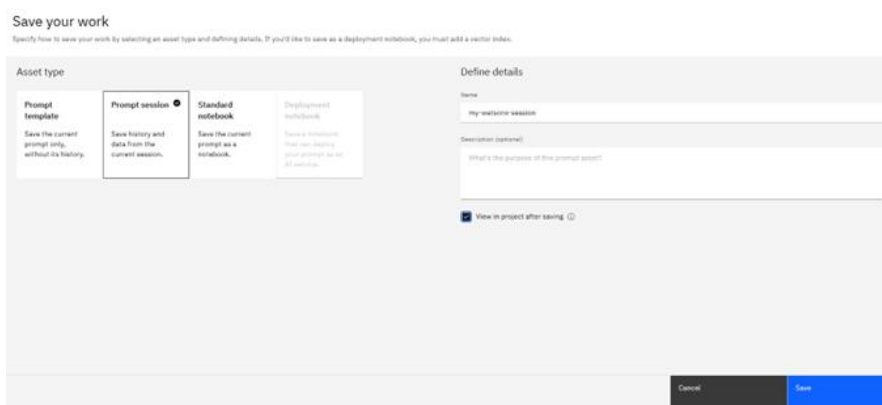
Save your Prompt Lab session

You can save your Prompt Lab editor session for later use.

1. At the top of the Prompt Lab screen, select the **Save work** dropdown button and then select the **Save as** option.



2. A **Save your work** widget will appear. Select **Prompt session** under the **Asset type** option.
3. Enter a **name** and check the **View in project after saving** option under the **Define details** section.
4. Finally, click the **Save** button. Once you save, you will see the saved work under the **Assets** tab



You can also save your work as:

- **Prompt template** to save only the current prompt without its history and selecting a **Task** suitable for your prompting.
- **Notebook** to continue prompting on a Jupyter Notebook environment. Prior knowledge of notebooks and Python programming language would be helpful to work with a Jupyter notebook. [Read more about notebooks.](#)

Save your work on watsonx.ai

Make sure to save any work you want to retain for your records. IBM Cloud accounts will be deactivated at the end of the hackathon. Follow the steps below to save your work:

1. Go to your project's 'Overview' tab.
2. Select the 'Export or import project' drop down below the Bell icon in the top menu bar.
3. Click the 'Export project' option. This will open 'Export project to desktop' screen.
4. Select all the assets shown in your project (Work saved as Project session cannot be exported) and click 'Export' on the bottom-right of the screen.

5. The next screen will ask for confirmation that all sensitive information has been removed.
6. Click on 'Continue export'.
7. The download (zip) will be initiated, and the file will be saved on your computer.

Appendix: Example use cases

You are not limited to these ideas, but here are several examples for how you could apply IBM Bob, with optional use of watsonx Orchestrate and watsonx.ai, to solve real challenges in this hackathon:

- **Smart developer onboarding assistant:** Help new developers understand and contribute to an unfamiliar codebase faster. Use IBM Bob to analyze repositories, explain architecture, generate setup guidance, and suggest starter tasks. Optionally use watsonx Orchestrate to automate onboarding activities and watsonx.ai to deliver conversational guidance.
- **Intelligent code review and quality coach:** Improve code quality and reduce review effort by using IBM Bob to analyze code changes, identify risks, explain findings, recommend fixes, and generate review summaries. Optionally use watsonx.ai to prioritize findings and watsonx Orchestrate to automate review workflows.
- **Automated testing and validation hub:** Reduce manual testing effort by using IBM Bob to generate unit tests, identify coverage gaps, validate code changes, and suggest improvements. Optionally use watsonx Orchestrate to orchestrate testing workflows and watsonx.ai to analyze test results and risk areas.
- **Release readiness and deployment assistant:** Help teams prepare software releases with confidence. Use IBM Bob to analyze code changes, review dependencies, summarize risks, generate release notes, and validate deployment requirements. Optionally use watsonx Orchestrate to automate release tasks and approvals.
- **Legacy application modernization accelerator:** Help teams understand, document, refactor, and modernize legacy applications. Use IBM Bob to explain existing code, identify modernization opportunities, generate updated components, and reduce migration effort. Optionally use watsonx.ai for impact analysis and watsonx Orchestrate to coordinate modernization workflows.